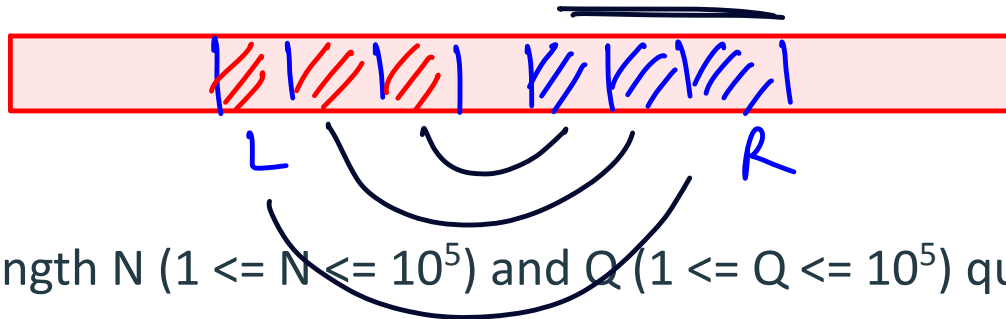




String Hashing Problem Solving

- Priyansh Agarwal

✓ Problem 1

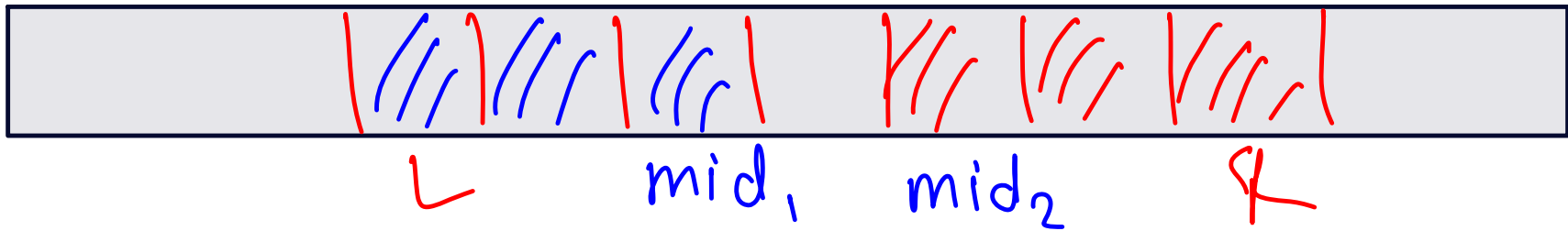


Given a string S of length N ($1 \leq N \leq 10^5$) and Q ($1 \leq Q \leq 10^5$) queries where each query contains 2 integers L and R ($1 \leq L \leq R \leq N$), find out for each query if the substring from L to R is a palindrome. ==

Example:

$$O(N + Q)$$

abccbd, $q_1 = [3, 4] \Rightarrow \text{Yes}$, $q_2 = [1, 5] \Rightarrow \text{No}$, $q_3 = [2, 4] \Rightarrow \text{Yes}$



$$h1 = s_0 \cdot p^0 + s_1 \cdot p^1 + s_2 \cdot p^2 + \dots + s_{n-1} \cdot p^{n-1}$$

$$h_1(i) = \left[\sum_{j=0}^i s_j \cdot p^j \right] \% m$$

$$\text{hash}(L \text{ to } mid_1) = \left[\frac{h_1[mid] - h_1[L-1]}{p^L} \right] \% m$$

$$h2 = s_{n-1} \cdot p^0 + s_{n-2} \cdot p^1 + \dots + s_0 \cdot p^{n-1}$$

$$h_2(i) = \sum_{j=n-1}^i s_j \cdot p^{n-j-1}$$

$$\text{hash}(L \text{ to } \text{mid}_2) = \frac{h_2(\text{mid}_2) - h_2(L-1)}{p^{n-L}}$$

$$h_2(n-1) = s_{n-1} \cdot p^0$$

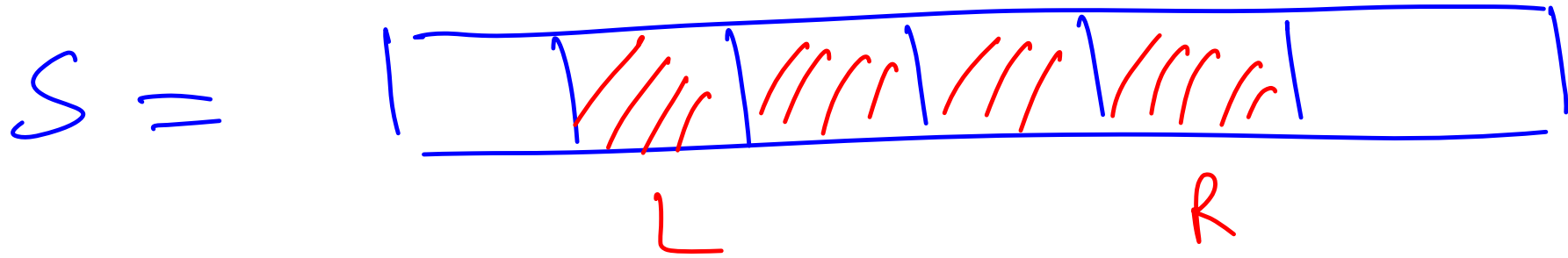
$$h_2(n-2) = s_{n-1} \cdot p^0 + s_{n-2} \cdot p^1$$

$$h_2(i) = S_{n-1} \cdot p^0 + S_{n-2} \cdot p^1 \\ \dots S_i \cdot p^{n-i-1}$$

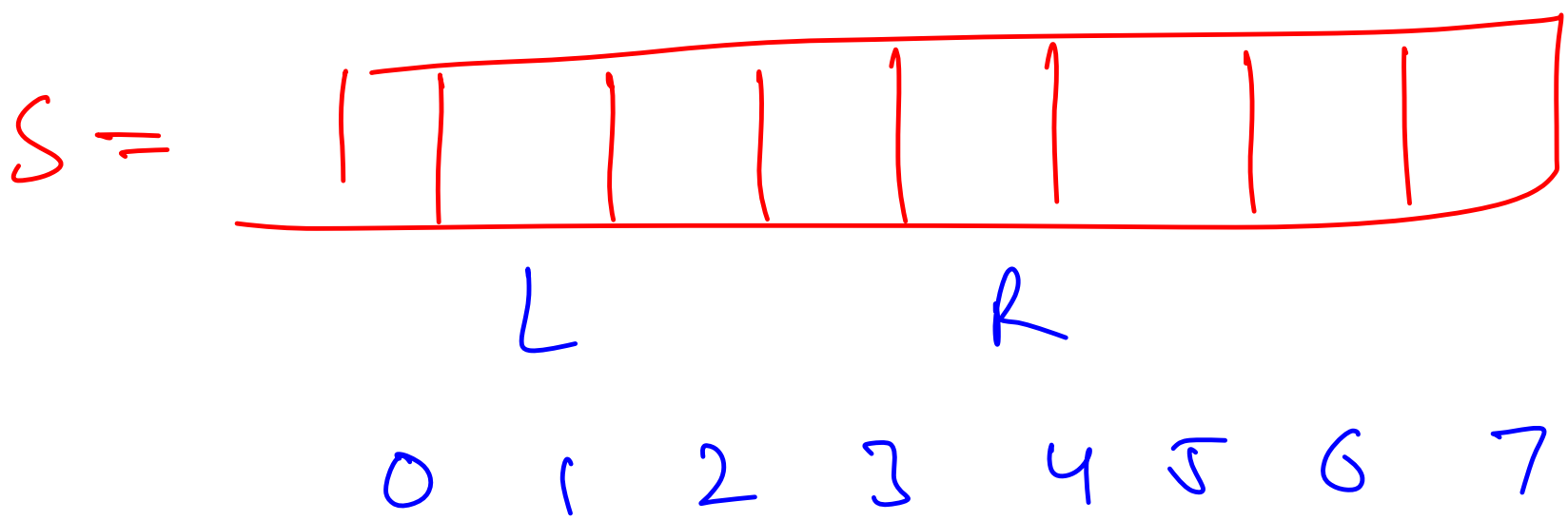
$$\text{hash}(R \text{ to } \text{mid}_2)$$

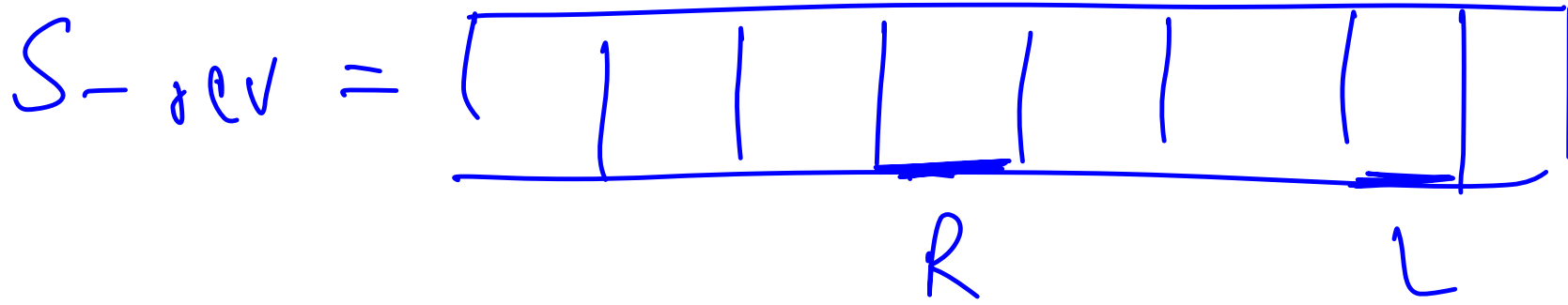
$$= S_R \cdot p^0 + S_{R-1} \cdot p^1 \dots$$

$$S_{\text{mid}_2} \cdot p^{R-\text{mid}_2}$$



$S(L \text{ to } R)$ must be same as
reverse of $S(L \text{ to } R)$





0
1
2
3
4
5
6
7

hash(L to R) for S

$$= \text{hash}(n-R-1 \text{ to } n-L-1)$$

for S_{-rev}



Problem 2

Given a string S of length N ($1 \leq N \leq 10^5$), find out the longest substring which occurs at least twice in the string.

Examples:

abcdabcd $\Rightarrow$ 4 (abcd)

abcd abcd

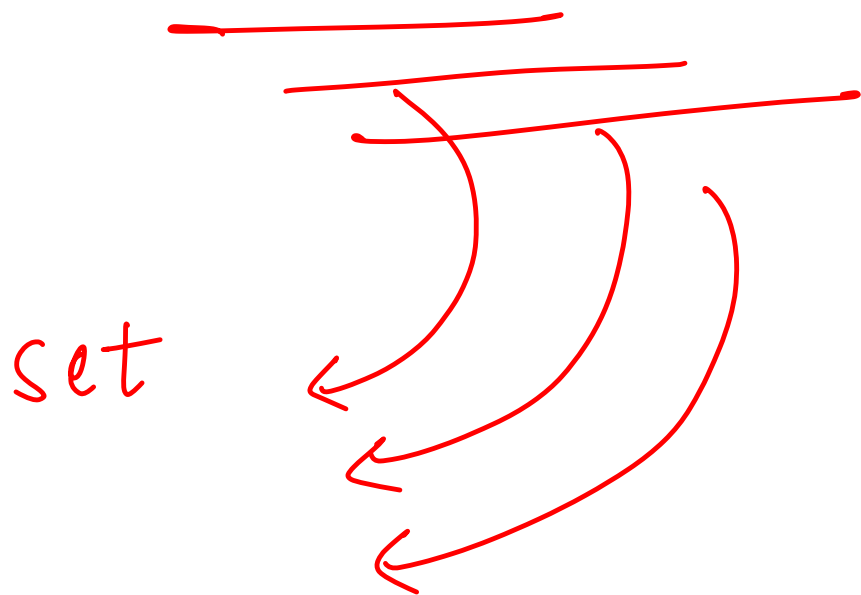
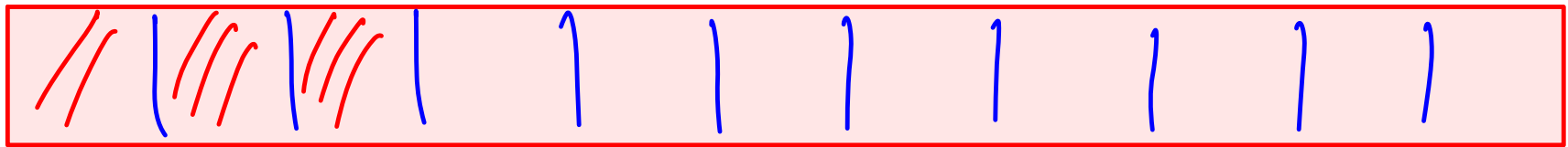
abbbbbbbd $\Rightarrow$ 5 (bbbbbb)

abbbbbbbd

abccab $\Rightarrow$ 2 (ab)

abccab

Can you quickly check if a substring
of length k repeats



$k=3$

no. of substring of
size $k = n - k + 1$

if any substring repeats
size of set $< n - k + 1$

$O(n \log n) \rightarrow \begin{cases} T \\ F \end{cases}$ if substring of size k repeats

abc

abc

ans = 3

T T T T T T T F F F F F F F F

x

Binary search on length $\rightarrow \log n$

predicate $\rightarrow n \log n$

$\Rightarrow O(n \log^2 n)$



Problem 3

Given a string S of length N ($1 \leq N \leq 10^5$), find out the longest palindromic substring.

Examples:

abbbbc $\Rightarrow$ 4 (bbbb)

abbabba $\Rightarrow$ 7 (abbabba)

abacde $\Rightarrow$ 3 (aba)

$len = k \Rightarrow$ is there a
↑ substring which is a
palindrome
 $(n - k + 1)$ ops
 $= \underline{O(n)}$

if a s len palindrome exists

then

3 ✓

1 ✓



$4 \leq x$

Binary search odd
 $\log n$

predicate $\rightarrow n$

best ans odd

Binary search on Even
 $\log n$

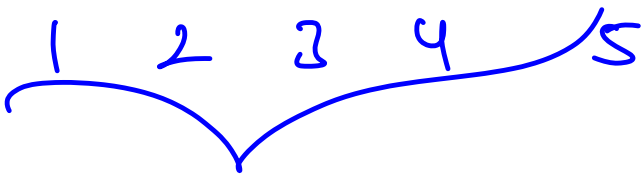
predicate $\rightarrow n$

best ans even

$n \log n$
 $\max (\quad)$

1	3	5	7	9	11
---	---	---	---	---	----

0 1 2 3 4 5



2	4	6	8	10
---	---	---	---	----

start = 0, end = 5

while (start ≤ end)

mid = (s + e) / 2

if (possible(odds[mid]))

ans = odds[mid]

s = m + 1

else

e = m - 1

Problem 4: Message Transmission Error



Link: <https://codeforces.com/contest/2010/problem/C2>

Obs 1 $\rightarrow$ ans must be a prefix of t
Obs 2 $\rightarrow$ ans must be a suffix of t

a c a c
a c a c a c
a c a c

ans must have length $> n/2$

$n = \text{length of } t$
ans = false

for (int $i = n/2 + 1, i \leq n; i++$)

if (prefix of len i = suffix of len i)

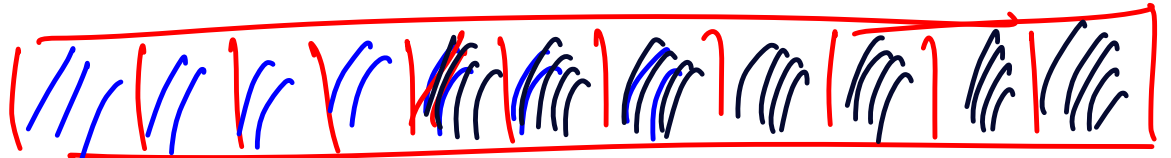
ans = true

break

↓
string hashing

$t = 11$

ans_len = 7



Problem 5: Minimal Rotation (CSES)



Link: <https://cses.fi/problemset/task/1110/>

Generate all cyclic shifts and compare them

shifts[n]

for (int i = 0 ; i < n ; i++) { $O(n)$

shifts[i] = generate_i_th_shift(i, s) $O(n)$

}

```

best-so-far = 0;
for (int i = 1; i < n; i++) { → O(n)
    if (shift[i] < shift[best-so-far] → O(n)
        best-so-far = i;
}

```

Visualizing i th cyclic shift

$S =$

a	b	c	d	e	f	g	h
---	---	---	---	---	---	---	---

 → 8

0 1 2 3 4 5 6 7

1st cyclic shift →

b	c	d	e	f	g	h	a
---	---	---	---	---	---	---	---

0 1 2 3 4 5 6 7

2nd cyclic shift

c	d	e	f	g	h	a	δ
0	1	2	3	4	5	6	7

j th character in i th rotation

$= [(i+j) \% n]$ th character in original

7th character in 2nd shift = δ

$= ((7+2) \% 8)$ th ch in original

let ~~obs~~ $shift[i][j] = shift[0][(i+j) \% n]$

$S \rightarrow$

a	b	c	d	e	f	g	h	i	j
0	1	2	3	4	5	6	7	8	9

4th shift

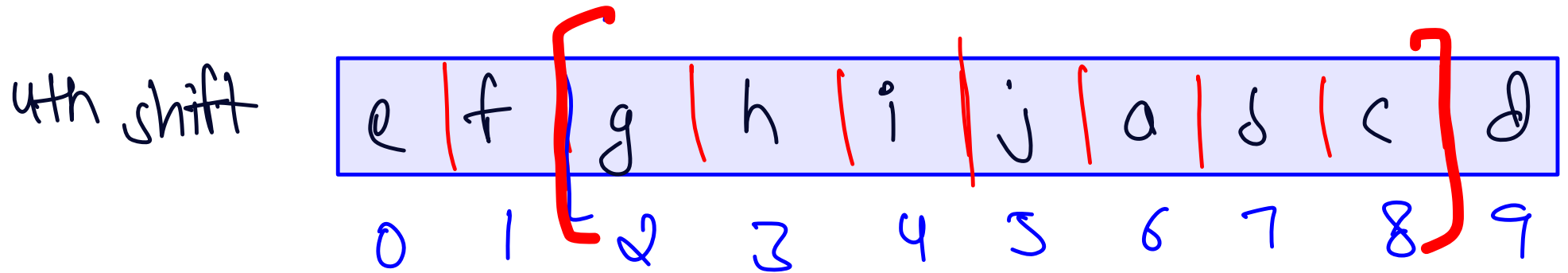
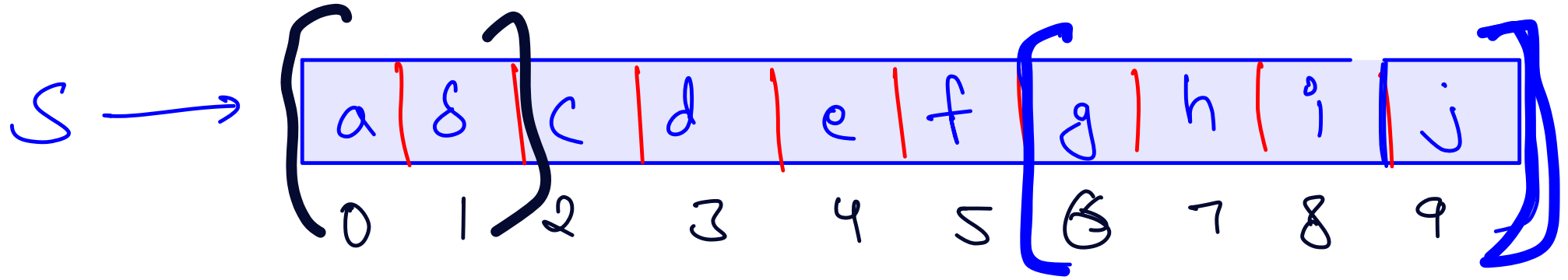
e	f	g	h	i	j	a	b	c	d
0	1	2	3	4	5	6	7	8	9

Substring from 2 to 4 in 4th shift

(L) (R) (i)

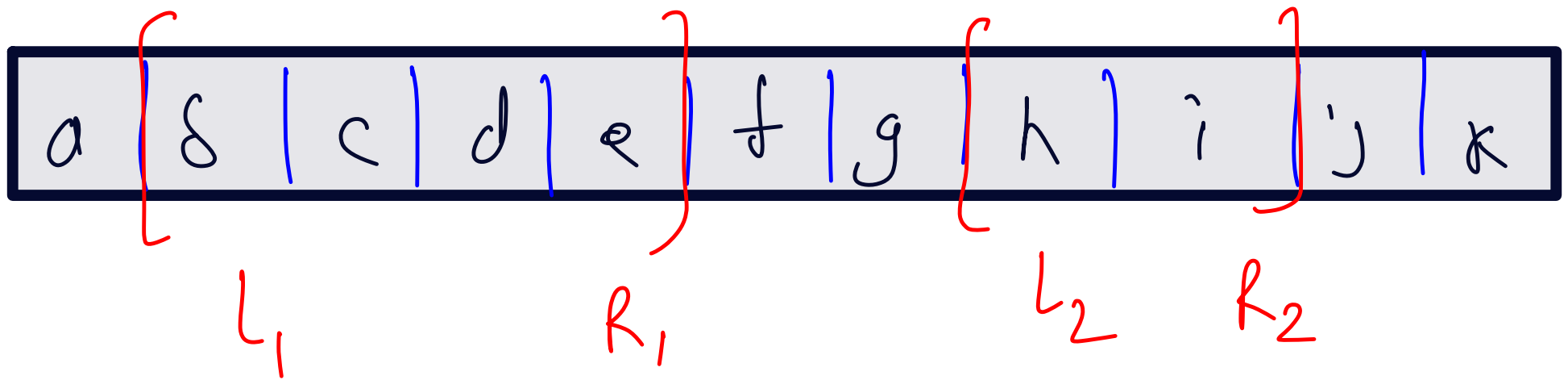
- [L+i] [R+i] in 0th shift

6 to 8 in original string



substring from 2 to 7 in 4th shift
 6 to 11 in original

hash $([6 \text{ to } 9] + [0 \text{ to } 1])$ in $O(1)$



$$\text{hash}(\overline{s[L_1 \text{ to } R_1]} + \overline{s[L_2 \text{ to } R_2]})$$

$$\text{hash}(s c d e h i)$$

$$\rightarrow = \text{hash}(L_1 \text{ to } R_1) + \text{hash}(L_2 \text{ to } R_2) \cdot p^{R_1 - L_1 + 1}$$

$$A = a_0 + a_1 + a_2$$

$$B = s_0 + s_1$$

$$\text{hash}(A + B)$$

$$= a_0 \cdot p^0 + a_1 \cdot p^1 + a_2 \cdot p^2 + s_0 \cdot p^3 + s_1 \cdot p^4$$

$$\begin{array}{c} \text{=} \\ \underbrace{\hspace{10em}} \\ \text{hash}(A) + p^3 \cdot \text{hash}(B) \end{array}$$

j th character in i th shift

= $[i+j] \% n$ th character in org string
 $O(1)$

d to r in i th shift

$d+i \leq n-1$ $d+i$ to $r+i$ in original string

$r+i > n-1$ $[d+i \text{ to } n-1] + [0 \text{ to } (r+i) \% n]$ in org
and

$d+i \leq n-1$ $O(1)$

$d+i > n-1 \rightarrow (d+i) \% n$ and $(r+i) \% n$

shift p

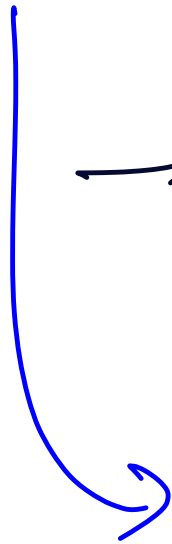
shift q

if (shift p < shift q)

1st char diff



binary search



binary search to find
first diff character

Problem 6: Strings Rotation



Link: https://acmp.ru/index.asp?main=task&id_task=829&locale=en